



COMPUTER SECURITY

2025

FINAL BANK

(Solved)



SENIOR
25

BY AB & AE & AM & MR

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Choose the correct answer:

1.	The need for <u>keeping information secret</u> .			
	a) Confidentiality	b) Integrity	c) Availability	d) None
2.	Trustworthiness of data and resources.			
	a) Confidentiality	b) Integrity	c) Availability	d) None
3.	The ability to use the information or resource			
	a) Confidentiality	b) Integrity	c) Availability	d) None
4.	Timely response is an expectation of _____ security goal			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
5.	_____ means that some unauthorized party (person, program) <u>has gained access to data</u>			
	a) Interception	b) Modification	c) Interruption	d) Fabrication
6.	_____ an asset of the system <i>becomes lost, <u>unavailable</u> or unusable</i>			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
7.	_____ means that some unauthorized party (person, program) has <u><i>gained access to an asset</i></u>			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
8.	_____ an additional computation, or modifying data being transmitted by the network.			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
9.	_____ counterfeit objects on a computing system. (adding records to an existing data base or insertion of spurious transactions to a network communication system.			
	a) Interruption	b) Interception	c) Modification	d) Fabrication
10.	Cryptography is used as a defense technique against the following attacks except _____			
	a) Modification	b) Interception	c) Interruption	d) Fabrication
11.	Erasure of program or data or malfunction of an OS file manager is considered _____			
	a) Modification	b) Interception	c) Interruption	d) Fabrication
12.	Flooding is an attack of _____			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
13.	Spoofing is an attack of _____			
	a) Integrity	b) Availability	c) Confidentiality	d) All are true
14.	_____ is when <u>one host pretends to be another</u>			
	a) Eavesdrop	b) Spoofing	c) Wiretap	d) Session Hijacking
15.	When a host pretends to be another , for example, using his IP's, is known as _____ attack			
	a) Eavesdrop	b) Masquerade	c) Wiretap	d) Session Hijacking
16.	_____ is intercepting and carrying on a session begun by another entity			
	a) Masquerade	b) Impersonation	c) Session Hijacking	d) Man-in-the middle



17.	_____ in which one entity intrudes between two others.							
	a)	Masquerade	b)	Impersonation	c) Session Hijacking	d)	Man-in-the-middle	
18.	A sender <u>should not be able falsely to deny later that he/she sent a message.</u>							
	a)	Integrity	b)	Confidently	c)	Privacy	d)	Non-repudiation
19.	Which security mechanism ensures that a user cannot deny their actions?							
	a)	Authentication	b)	Authorization	c)	Confidentiality	d)	Non-repudiation
20.	Program controls include:							
	a)	Peer review	b)	Proof of program correctness	c)	Independent testing	d)	All
21.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
22.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
23.	Eavesdropping and impersonate are attacks of _____ Respectively							
	a)	Availability , Confidentiality	b)	Availability , Integrity	c)	Confidentiality , Integrity	d)	Integrity , Confidentiality
24.	Choose the correct sentence?							
	a)	Erasure of a program is considered an interruption threat						
	b)	TOCTOU is considered an attack of availability						
	c)	Flooding are attacks of confidentiality						
	d)	Error is a mistake in interpreting a syntax error in a piece of code						
25.	_____ Consuming processor capacity, system memory , disk space							
	a)	Bacteria	b)	Resident	c)	Virus	d)	Worm
26.	_____ <u>Attaches itself to program</u> and propagates copies of itself to other programs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
27.	_____ Contains <u>unexpected</u> , additional functionality							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
28.	_____ Triggers action when <u>condition</u> occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
29.	_____ Triggers action when <u>specified time</u> occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
30.	_____ Allows unauthorized access to functionality							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
31.	_____ Propagates copies of itself through a network							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit



32.	_____ Replicates itself without limit to exhaust resources			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
33.	_____ is a class of malicious code that " detonates " or goes off when a specified condition occurs			
	a) Virus	b) Trojan horse	c) Logic bomb	d) Time bomb
34.	The _____ is a code embedded in program that is set to explode when certain condition is/are met			
	a) Trojan house	b) Trapdoor	c) Worm	d) Logical bomb
35.	_____ replicates a copy of itself to other systems to execute a malicious code			
	a) Virus	b) Worm	c) Trojan horse	d) Trapdoor
36.	_____ is a program that spreads copies of itself through a network			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
37.	_____ is a feature in a program by which someone can access the program <u>other than by the obvious, direct call</u> , perhaps with special privileges			
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit
38.	_____ virus runs when its attached program executes and terminates when its attached program ends			
	a) Transient	b) Resident	c) Virus	d) Worm
39.	_____ virus locates itself in memory so that it can <u>remain active</u> even after its attached program ends			
	a) Transient	b) Resident	c) Virus	d) Worm
40.	A type of virus that executes only once , spreading its infection and causing its effect in that one execution .			
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus
41.	A kind of malicious software runs when a computer is started , it changes control on the OS instructions			
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus
42.	A type of virus which is active as long as the computer running			
	a) Memory resident virus	b) Brain virus	c) Polymorphic virus	d) Logic bomb
43.	Virus is active as long as the computer running			
	a) Memory resident	b) Brain virus	c) Polymorphic virus	d) Trojan horse
44.	Viruses resident in application programs and libraries are categorized as:			
	a) One time execution virus	b) Boot sector virus	c) Other homes of virus	d) Memory resident
45.	A program virus attaches itself to a program ; then, whenever the program is run, the virus is activated. This kind of attachment is usually known as _____			
	a) integrated virus	b) Surrounded virus	c) Appended virus	d) replacement virus
46.	controls against program threats include the following actions except _____			
	a) peer review process	b) Independent testing	c) Proof of correctness	d) Decapsulation of different process



47.	Each virus has a signature that could be used to detect its existence, such as:				
	a)	Its location	b)	its execution performance	c) its replication technique
48.	What is a weakness of Caesar cipher?				
	a)	Vulnerable to brute-force attacks	b)	Vulnerable to frequency analysis	c) limited key space
49.	What is the main weakness of <u>simple substitution</u> ciphers?				
	a)	They are slow to encrypt messages	b)	They require large keys	c) They can be easily broken using frequency analysis
50.	What is true about Steganography?				
	a)	it is used to encrypt images and send it in unreadable forms			
	b)	The encrypted file has smaller size compared with the original one			
	c)	it can be broken by brute-force attack			
	d)	it uses images to hide important data in LSB of each image's pixel			
51.	<u>Steganography</u> uses _____ in graphic image to hide important data				
	a)	LSB	b)	HSB	c) MSB
52.	_____ is a mistake in <i>interpreting a requirement</i> or a syntax error in a piece of code				
	a)	Error	b)	Flaws	c) Bugs
53.	_____ is a human mistake in performing some software activity that <u>may lead to a fault</u> in computer program				
	a)	Error	b)	Flaws	c) Bugs
54.	Validation error , domain error , and boundary condition violation are types of _____				
	a)	Error	b)	Flaws	c) Bugs
55.	A _____ <u>may cause a failure</u> (which is a departure from the system's required behavior).				
	a)	Bug	b)	Error	c) fault
56.	A _____ is an inside view of the system , <u>seen by the developers</u> , whereas a _____ is an outside view <u>seen by the user</u>				
	a)	Failure, fault	b)	Fault, Failure	c) Error, Bug
57.	What is true about program <u>flaws</u> ?				
	a)	It could be validation error or domain error			
	b)	It could lead to failure			
	c)	It could be malicious or non malicious			
	d)	All are true			
58.	Consist of replacing the valid source and/or destination IP address with false ones				
	a)	Smurf attack	b)	Spoofing attack	c) DoS



59.	For exchanging symmetric key among 10 user , we need _____ exchange operation $n = 10, \frac{n \times n - 1}{2} = \frac{10 \times 9}{2} = 45$			
	a) 45	b) 100	c) 90	d) 1000
60.	Authentication based on <u>Something the individual has</u> as:			
	a) ID	b) Password	c) ATM card	d) PIN
61.	Authentication based on <u>Something the individual knows</u> as:			
	a) PIN	b) Password	c) ATM card	d) A & B
62.	Authentication based on <u>Something the individual is</u> as:			
	a) Voice patterns	b) Handwriting	c) Fingerprint	d) All
63.	Password and PIN are examples for user authentication of something _____			
	a) The user has	b) The user is	c) The user knows	d) The user
64.	What is the Row Transposition Cipher?			
	a) A substitution cipher that rearranges the order of letters in a message			
	b) A cipher that replaces each letter in a message with a numerical value			
	c) A cipher that applies a mathematical formula to each character in a message			
	d) A cipher that rearranges the order of words in a message			
65.	Which of the following is a common technique used with <u>transposition</u> cipher?			
	a) Rail fence	b) Caesar	c) Vagenre	d) Playfair
66.	The mechanisms used to confirm a user's identity is known as:			
	a) Identification	b) Authentication	c) Authorization	d) Access limitation
67.	Each virus has a signature that could be used to detect its existence, such as:			
	a) Location	b) Execution performance	c) Spreading	d) All are true
68.	In _____, Users interact with directory to obtain any desired public key securely			
	a) Certificate authority	b) Public Key authority	c) Public announcement	d) Public Discovery
69.	_____ allow key exchange without real-time access to public-key authority			
	a) Certificate authority	b) Public key authority	c) Public announcement	d) Public directory
70.	Example for <u>classical</u> substitution encryption			
	a) Playfair	b) DES	c) 3DES	d) AES
71.	Triple-DES with two keys is defined by $C = EK_3(DK_2(EK_1(P)))$, if K1 equal K3 , the triple-DES can be considered هنا هو بيقولك ان k1 يساوي k3، وأول خطوه حصلت انه شفر ال P ب k1 وبعدين تاني خطوة فك تشفيرها ب k2 بس ده مش هيرجع لك P الاصلية لان k1 مش هي k2 فدي هتعتبر خطوة تشفير تانية، وبعدين قام مشفر ب k3 فكدا حصل تشفير ثلاث مرات يبقى ده triple DES			
	a) Double DES	b) Triple-DES	c) Single-DES	d) Useless encryption



72.	Triple-DES with two keys is defined by $C = Ek_1(Dk_2(Ek_1(P)))$, if K1 equals K2 , the triple-DES will be هنا هو بيقولك ان k1 يساوي k2، وأول خطوه حصلت انه شغل ال m ب k1 وبعدين ثاني خطوه فك تشفيرها ب k2 فهيرجع ال الأصلية عادي يعني اول خطوه اتلغت بتاني خطوه، وبعدين رجع شغل ب k1 فيبقى كأنه عمل تشفير مره واحده بس فوهربقى single DES			
	a) Double DES	b) Triple-DES	c) Single-DES	d) Useless encryption
73.	Which of the following is a good description for Triple-DES ?			
	a) $C = Ek_1(Dk_2(Ek_1(P)))$	b) $C = Ek_2(Dk_1(Ek_1(P)))$	c) $C = Ek_3(Ek_2(Ek_1(P)))$	d) $C = Dk_3(Dk_2(Dk_1(P)))$
74.	DES encrypt data in blocks of size _____, using a key of length _____			
	a) 64-bit , 32-bit	b) 32-bit, 32-bit	c) 32-bit , 56-bit	d) 64-bit , 56-bit
75.	What is the number of rounds in <u>single DES</u> ?			
	a) 64	b) 56	c) 48	d) 16
76.	What is the block size of <u>triple DES</u> ?			
	a) 64	b) 128	c) 256	d) 512
77.	_____ is an attempt to discover PW by attempting to use every possible PW <u>from a predefined list</u> .			
	a) Brute-force attack	b) Dictionary attack	c) Sniffing attack	d) Cracking
78.	_____ is a type of attack that depends on analyzing relations among the contents on cipher text			
	a) Time attack	b) Statistical analysis	c) Dictionary attack	d) Brute force
79.	_____ provides a centralized authentication server to authenticate users to servers and servers to users .			
	a) Trusted Third Party (TTP)	b) Trusted Peer Party (TPP)	c) Trusted Party Model (TPM)	d) None
80.	_____ is a protocol that <u>produces the same effect as a real signature</u>			
	a) Digital signatures	b) Symmetric Key	c) Asymmetric Key	d) None
81.	Digital signatures provide the ability to:			
	a) verify author, date & time of signature	b) be verified by third parties to resolve disputes	c) authenticate message contents	d) All are true
82.	_____ refers to a key storage facility which provides secure storage of keys for future use, e.g.			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery
83.	_____ refers to procedures by which keys for notarization or nonrepudiation services can be securely archived .			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery
84.	_____ enables parties to update securely their keying material.			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery



85.	_____ refers to cryptographic keys which may become lost due to human error, software bugs, or hardware malfunction				
	a)	Storage of Keying Material	b)	Key Archival	c) Key Replacement d) Key Recovery
86.	_____ refers to procedures by which parties are assured of the secure destruction of keys that are no longer needed				
	a)	Storage of Keying Material	b)	Key Archival	c) Key Replacement d) Key Deletion
87.	_____ refers to the procedure by which keys or pairs of keys of good cryptographic quality are securely and unpredictably generated				
	a)	Storage of Keying Material	b)	Key Archival	c) Key Replacement d) Key Generation
88.	_____ provide some link between an entity and its uniquely defined keys				
	a)	Entity Registration	b)	Key Replacement	c) Entity Authentication d) Entity Replacement
89.	_____ is to corroborate that an entity is the one claimed				
	a)	Key Replacement	c)	Entity Authentication	d) Entity Replacement d) Entity Registration
90.	_____ is the registration of messages to attest at a later stage its content origin .				
	a)	Notarization	b)	Proof	c) Nonrepudiation of Reception d) none
91.	_____ secret keys and possibly other data are to be kept confidential while transmitted or stored				
	a)	Data Confidentiality	b)	Modification Detection	c) Replay Detection d) Timeliness
92.	_____ is to counter the active threat of unauthorized modification of data items				
	a)	Data Confidentiality	b)	Modification Detection	c) Replay Detection d) Timeliness
93.	_____ is to counter unauthorized duplication of data items.				
	a)	Data Confidentiality	b)	Modification Detection	c) Replay Detection d) Timeliness
94.	_____ is required that the response to a challenge message is prompt and does not allow for playback of some authentic response message by an impersonator.				
	a)	Data Confidentiality	b)	Modification Detection	c) Replay Detection d) Timeliness
95.	_____ It involves storing, safeguarding, and activating keys				
	a)	key management	b)	key distribution	c) Key Replacement d) Key Recovery
96.	_____ refers to the procedures by which keys are securely provided to parties legitimately asking for them .				
	a)	key management	b)	key distribution	c) Key Replacement d) Key Recovery
97.	The _____ private key is used to decrypt cipher text in asymmetric cryptosystem				
	a)	Sender	b)	Receiver	c) Both A & B d) None
98.	The _____ key is used to encrypt plain text in asymmetric cryptosystem				
	a)	public	b)	private	c) Both A & B d) None



99.	_____ is the science of code breaking in a cryptographic schema			
	a) Cryptography	b) Cryptanalysis	c) Cryptology	d) None
100.	_____ is the study of encryption principles and methods.			
	a) Cryptography	b) Cryptanalysis	c) Cryptology	d) None
101.	_____ is the field of both cryptography and cryptoanalysis.			
	a) Cryptography	b) Cryptanalysis	c) Cryptology	d) None
102.	In cryptography system, _____ is the original message before being transmitted			
	a) Cipher text	b) encrypted	c) Plain text	d) None
103.	In Caesar cipher, shifting a letter by +3 positions means that			
	a) A becomes D, B becomes E, C becomes F, and so on			
	b) A becomes X, B becomes Y, C becomes Z, and so on			
	c) A becomes C, B becomes F, C becomes I, and so on			
	d) A becomes B, B becomes C, C becomes D, and so on			
104.	Type of firewall can decide what services to perform on the user's behalf in accordance with its available knowledge			
	a) Screening routers	b) Stateful inspection	c) Guards	d) Application proxies
105.	It perform a simple pattern-matching to report situations that corresponding to a known attack type			
	a) Normal based	b) Signature-based IDS	c) Heuristic IDS	d) Honey pot
106.	It implies overhearing without expending any extra effort			
	a) Eavesdrop	b) Wiretap	c) Spoofing	d) Session hijacking
107.	Triple-DES with two keys id defined by $C= Ek_3(Dk_2(Ek_1(P)))$, if K_1 equal K_3 , the triple-DES can be considered			
	a) Double DES	b) Triple DES	c) Single DES	d) Useless encryption
108.	Which of the following isn't a design goal of firewalls			
	a) Always invoked	b) Tamper-proof	c) Complex code	d) Simple functionality
109.	It tracks the sequence of packets and conditions from one packet to another to thwart an attack			
	a) Screening routers	b) Stateful inspection	c) Application proxies	d) Guards
110.	A sender should not be able falsely to deny later that he/she sent a message			
	a) Integrity	b) Confidentiality	c) Privacy	d) Non-repudiation
111.	Physical OS separation approach _____ the resource sharing and _____ the security level			
	a) Minimizes , Maximizes	b) Maximizes , Maximizes	c) Maximizes , Minimizes	d) Minimizes , Minimizes
112.	Physical OS separation approach _____ the security level and _____ the resource sharing			
	a) Minimizes , Maximizes	b) Maximizes , Minimizes	c) Minimizes, Minimizes	d) Maximizes , Maximizes
113.	Kerberos system is considered _____ party _____ way authentication technique			
	a) Two , One	b) Two , Two	c) Three , Two	d) Three , One



114.	<u>Digital signature</u> provides the following characteristics expect _____							
	a)	Message authentication	b)	Receiver authentication	c)	Sender authentication	d)	Message encryption
115.	_____ Is used for managing audit trails and <u>highlighting user violation of policy or normal activity</u>							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
116.	Correction system configuration error is a function of _____							
	a)	Heuristic-based IDS	b)	Personal firewall	c)	Proxy	d)	Signature-based IDS
117.	Denning's IDS model builds heuristic decision model based on tracing _____							
	a)	Actions and it's initiator	b)	Timestamp of each action	c)	Exception condition of each action	d)	All
118.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
119.	_____ means that some unauthorized party (person, program) has <u>gained access to data</u>							
	a)	Interception	b)	Modification	c)	Interruption	d)	Fabrication
120.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
121.	A virus Is designed to <u>hide itself from detection</u> by antivirus software							
	a)	Brain virus	b)	Boot sector virus	c)	Polymorphic virus	d)	Trojan horse
122.	_____ record an intruder's actions and even attempt to trace who the attacker is from actions, packet data, or connection							
	a)	ISS scanner	b)	Nessus	c)	Honey pot	d)	tripwire
123.	Which of the following is/are considered a candidate IDS profile?							
	a)	Logon and session activity	b)	Command or program usage	c)	File access activity	d)	All
124.	_____ used to <u>protect the networks hosts from an outside address</u>							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
125.	Denning's IDS model is example of _____							
	a)	Host based IDS	b)	Signature based IDS	c)	Anomaly based IDS	d)	Hybrid IDS
126.	In which different processes use different physical objects, such as separate printers for output requiring different levels of security							
	a)	Temporal separation	b)	Logical separation	c)	Physical separation	d)	Cryptographic separation
127.	Steganography uses _____ in graphic image <u>to hide important data</u>							
	a)	LSB	b)	HSB	c)	MSB	d)	NSB
128.	_____ Controls access on the basis of packet address or <u>specific transport protocol type</u>							
	a)	Stateful inspection	b)	Screening routers	c)	Application proxies	d)	Guards



129.	A firewall system characterized by being two-headed device is							
	a)	Screening routers	b)	Stateful inspection	c)	Application proxies	d)	Guards
130.	<u> </u> Builds a model of acceptable behavior and <u>flag exceptions to that model</u>							
	a)	Normal based	b)	Signature-based IDS	c)	Heuristic IDS	d)	Honey pot
131.	It attracts the intruder , record his actions, and even attempt to trace who the attacker is from action, packet data, or connections.							
	a)	Heuristic IDS	b)	Honey pot	c)	Normal Based	d)	Signature-based IDS
132.	Intercepting and <u>carrying on a session begun by another entity</u> is known as <u> </u> attack							
	a)	Eavesdrop	b)	Wiretap	c)	Spoofing	d)	Session hijacking
133.	<u> </u> this from of protection <u>limits not just the access to an object but the use made of that object</u> after it has been access							
	a)	Share via access limitation	b)	Share all or share nothing	c)	Share Via capabilities	d)	Limit use of an object
134.	<u> </u> is when one host pretends to be another.							
	a)	Eavesdrop	b)	Spoofing	c)	Wiretap	d)	Session hijacking
135.	With this form of protection, the owner of an object declares it to be public or private							
	a)	Share via access limitation	b)	Share all or share nothing	c)	Don't protect model	d)	Isolate mode
136.	<u> </u> in which processes conceal their data and computations in such a way that they are unintelligible to outside processes							
	a)	Temporal Separation	b)	Logical Separation	c)	Physical separation	d)	Cryptographic separation
137.	<u> </u> Change control on the OS instructions, when a computer is started							
	a)	Memory resident virus	b)	Boot sector virus	c)	One time execution	d)	All
138.	When <u>OS runs the applications</u> of different security requirements <u>under illusion that no other processes exist</u>							
	a)	Logical separation	b)	Physical separation	c)	Temporal separation	d)	Space separation
139.	<u> </u> perform simple pattern-matching based on logon and session activity to detect attack							
	a)	Heuristic -based IDS	b)	Personal Firewall	c)	Proxy	d)	Signature-based IDS
140.	Firewall is an executable code runs on a dedicated device to protect inside from outside networks							
	a)	Firewall	b)	IDS	c)	Antivirus	d)	Virus scanner
141.	A company wants to <u>block the connection between its LAN and "youtube "</u> , simply they can apply <u> </u>							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
142.	A company wants to set up an online price list so that outsiders can see the products and prices offered. It wants to be sure that no outsider can change the prices or product list, the outsiders can access only the price list, not any of the more sensitive files stored inside.							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls



143.	A company wants to allow its employees to <u>use e-mail up to a limit of so many messages</u> or so many characters of e-mail in the last so many days, simply they can apply _____							
	a)	Packet filtering gateways	b)	Application proxies firewall	c)	Stateful inspection firewalls	d)	Guards firewalls
144.	Which of the following is best description of Stateful inspection firewall working principle?							
	a)	It can see address and services protocol type	b)	It can see full data portion of packet	c)	It can see either address or data	d)	It can see full text of communication
145.	What is true about IDS?							
	a)	it could be stand-alone device attached to the network to monitor traffic throughout that network						
	b)	Honey pot is considered an example for IDS						
	c)	Regarding anomaly-based IDS, admin can mark a flagged behavior as acceptable or not.						
	d)	All						
146.	_____ means intercepting and <u>carrying on a session</u> begun by another entity.							
	a)	Masquerade	b)	Man-in-the-Middle Attack	c)	Session hijacking	d)	Impersonation
147.	Which of the following can describe the function of Firewall system ?							
	a)	To encrypt all network traffic to prevent eavesdropping						
	b)	To block all incoming network traffic except for authorized sources						
	c)	To prevent all unauthorized activity of networks' internal users						
	d)	To monitor internal network traffic and detect potential security breaches						
148.	Denning's model is an example for _____							
	a)	heuristic based IDS	b)	signature based IDS	c)	Virus Scan software	d)	Firewall model
149.	Which profile is recommended for auditing?							
	a)	logon and session activity	b)	command or program usage	c)	file access activity	d)	All
150.	When OS runs different process of different security requirements on the same time with shared resources under illusion that no other processes exist							
	a)	Cryptographic separation	b)	Temporal separation	c)	logical separation	d)	Physical separation
151.	if different processes can run concurrently <u>unaware of the presence of each other</u> , and each process has its own address space, files, and other objects. This OS level of sharing is categorized as:							
	a)	Isolation level	b)	Share via capabilities	c)	Share via access limitations	d)	Share all or share nothing
152.	What is TRUE about Kerberos authentication protocol?							
	a)	It is a two-party, two-way authentication protocol						
	b)	It is a two-party, three-way authentication protocol						
	c)	It is a three-party, two-way authentication protocol						
	d)	It is a three-party, three-way authentication protocol						



153.	What is FALSE about Kerberos authentication protocol?						
	a)	It is a two-party one-way authentication protocol					
	b)	It requires two special function servers					
	c)	It is always a third party					
	d)	It used symmetric key encryption					
154.	The ticket issued to the user by Kerberos protocol includes the following data						
	a)	User login details	b)	Time-date stamp	c)	User privilege on the accessed application	d)
155.	What is a common example of a man-in-the-middle attack						
	a)	Trojan horse	b)	DDoS attack	c)	Wi-Fi eavesdropping	d)
156.	What is true about IDS?						
	a)	It could be stand-alone device attached to the network to monitor traffic throughout that network					
	b)	Honey pot is considered an example for IDS					
	c)	Regarding anomaly-based IDS, admin can mark a flagged behavior as acceptable or not					
	d)	All					
157.	Which of the following is best description of stateful inspection firewall working principle?						
	a)	It can see address and services protocol type					
	b)	it can see either address or data					
	c)	it can see full portion of packet					
	d)	It can see full text of communication					
158.	Which of the following best describes a Denial-of-service (DoS: attack)?						
	a)	An attack that locks users out of their accounts					
	b)	An attack that forces a system to shut down by overwhelming it with traffic					
	c)	An attack that steals sensitive information					
	d)	An attack that exploits vulnerabilities in cryptographic algorithms					
159.	The Feistel function F-function: in DES Includes which major operation? Correct: Expansion → XOR with round key → Substitution → Permutation						
	a)	Substitution, Expansion, Permutation, XOR with round key					
	b)	Expansion, Substitution, Permutation, XOR with round key					
	c)	Expansion, Permutation, Substitution, XOR with round key					
	d)	XOR with round key, Expansion, Substitution, Permutation					



160.	What is a disadvantage of Rail Fence cipher compared to other transposition ciphers?
	a) it is more complex to implement
	b) it has a smaller key space
	c) it is slower in encrypting and decrypting
	d) It is more vulnerable to attacks
161.	What distinguishes <u>a trojan horse from a virus</u> ?
	a) Trojans cannot execute on their own
	b) Trojans spread through email attachments, while viruses spread via networks
	c) Trojans define themselves as legitimate software, while viruses attach to files
	d) Trojans self-replicate, whereas viruses do not
162.	What is the <u>purpose of S-boxes in DES</u> ?
	a) To compress data
	b) To provide confusion by replacing input bits with different bits
	c) To generate subkeys
	d) To provide permutation
163.	Which of the following best describes how <u>transposition ciphers</u> provide security?
	a) By completely replacing the original letter with different characters
	b) By making frequency analysis difficult through letter rearrangement
	c) By requiring a secret key that cannot be guessed
	d) By using a complex mathematical function to generate ciphertext
164.	Which of the following statements about <u>transposition ciphers</u> is true ?
	a) They require a key that determines letter substitution
	b) They are always more secure than substitution ciphers
	c) They change the letters in the plaintext
	d) They keep the original plaintext letters but rearrange their order
165.	What is the main purpose of <u>encryption</u> in computer security?
	a) To protect data from unauthorized access
	b) To speed up data transmission
	c) To create backup copies of data
	d) To hide system vulnerabilities
166.	What is <u>buffer overflow</u> ?
	a) A type of firewall
	b) A vulnerability where data overflows into adjacent memory locations
	c) A secure threat for data storage
	d) A type of ransomware attack



167.	Which of the following is NOT an encryption algorithm?			
	a) RSA	b) AES	c) DSS	d) DES
168.	How many subkeys are generated from the main key in DES? (how many rounds)			
	a) 8	b) 16	c) 32	d) 10
169.	What type of attack involves <i>intercepting and modifying communication</i> between two parties without their knowledge?			
	a) Phishing	b) Denial-of-Service	c) Brute-Force	d) Man-in-the-middle
170.	What is a security threat?			
	a) A program that prevents hacking attempts			
	b) Any event or action that could cause damage to a system or data			
	c) A measure used to warn user			
	d) Any action that improves system security			
171.	_____ runs on a <i>single workstation or client or host</i> , to protect that one host			
	a) Signature-based IDS	b) Anomaly-based IDS	c) Host-based IDS	d) Network-based IDS
172.	_____ is a stand-alone device attached to the network to <i>monitor traffic throughout that network</i>			
	a) Signature-based IDS	b) Anomaly-based IDS	c) Host-based IDS	d) Network-based IDS
173.	Which type of IDS Monitors for deviations from normal network behavior and <i>can detect previously unknown threats</i> ?			
	a) Signature-based IDS	b) Anomaly-based IDS (Heuristic IDS)	c) Host-based IDS	d) Network-based IDS
174.	_____ perform simple pattern-matching based on logon and session activity to detect attack			
	a) Heuristic -based IDS	b) Personal Firewall	c) Proxy	d) Signature-based IDS
175.	What is the difference between a network-based IDS and host-based IDS?			
	a) A network-based IDS monitors network traffic while a host-based IDS monitors system logs			
	b) A network-based IDS monitors system logs while a host-based IDS monitors network traffic			
	c) A network-based IDS is installed on individual hosts while a host-based IDS monitors network traffic			
	d) A network-based IDS is installed on the network while a host-based IDS is installed on individual hosts			
176.	Which type of firewall intercepts all communication between an internal network and the internet ?			
	a) Packet-filtering firewall	b) Application firewall	c) Stateful inspection firewall	d) Proxy firewall
177.	Which type of firewall can be used to control access to packets on the basis of packet address ?			
	a) Packet-filtering firewall	b) Application firewall	c) Stateful inspection firewall	d) Proxy firewall



True/False:

1.	<u>Spoofing</u> attack consists of <u>replacing the valid</u> source and/or destination IP address <u>with false one</u>	T
2.	<u>Bug</u> is a <u>mistake in interpreting a requirement</u> or <u>syntax error in a piece of code</u>	T
3.	For exchanging symmetric key among 10 users, we need 100 exchange operation (45)	F
4.	Erasure of program or data or malfunction of an OS file manager is considered fabrication (interruption)	F
5.	<u>Validation error</u> , <u>domain error</u> , and <u>boundary condition violation</u> are types of <u>flaws</u>	T
6.	Erasure of program or data or malfunction of an OS file manager is considered Modification (interruption)	F
7.	<u>Timely response</u> is an expectation of <u>availability</u> security goal.	T
8.	<u>Verification of the user's identity</u> is known as authorization process (authentication)	F
9.	Digital signatures provide the ability to verify both owner and content of message	T
10.	The plain text must be broken into a block size of 128bit as an input of <u>Triple DES</u> (64-bit)	F
11.	<u>Polymorphic virus</u> is a form of virus explicitly designed to <u>hide itself from detection</u>	T
12.	A <u>boot sector</u> virus changes control on the OS instructions, when a computer <u>is started</u>	T
13.	Playfair cipher rearranges the order of words in a message (letters)	F
14.	Shift cipher applies a <u>mathematical formula</u> to each letter in a message that replaces each letter with a numerical value. (Vigenère)	F
15.	<u>Flooding</u> is an attack of <u>availability</u>	T
16.	Authorization is mechanism used to confirm a user's identity (authentication)	F
17.	Fingerprint is used to verify the user as an example for something <u>user has</u> (ATM card)	F
18.	Cryptographic separation is applied when OS runs different process of different security requirements on different time with separate resources (temporal)	F
19.	Trapdoor is a malicious software runs when a computer is started, it Changes control on the OS instructions (boot sector virus)	F
20.	Program controls must include <u>peer review steps</u>	T
21.	In context of Program controls, <u>proof of program correctness</u> is done based on <u>check list</u>	T
22.	<u>Code flaws</u> include validation error or domain error	T
23.	Heuristic IDS performs simple pattern-matching to report situation that corresponding to a known attack type (signature-based)	F
24.	<u>Eavesdrop</u> Implies overhearing <u>without expending any extra effort</u>	T
25.	The firewall can't be able to detect the signature of the attack if it was broken into multiple packets each have very short length	T
26.	A personal firewall is an application program that runs on a user's workstation usually under his policy	T
27.	Auditing is difficult when applying Packet filtering firewalls	T
28.	A <u>network-based IDS</u> is installed <u>on the network</u> while a <u>host-based IDS</u> is installed on <u>individual hosts</u> .	T



MCQ CASES

Case 1	A library wants to make a available certain documents but, to support fair use of copyrighted matter, it will allow a user to retrieve only the first so many characters of a document, after that amount, the library will require the user to pay a fee that will be forwarded to the author			
Case 2	A university wants to allow its students to use <u>e-mail</u> up to a limit of so many messages or so many characters of e-mail in the last so many days, although this result could be achieved by modifying e-mail handlers, it is more easily done by monitoring the common point through which all e-mail flows, the <u>e-mail transfer protocol</u>			
Case 3	A company wants to allow dial-in access by its employees, without exposing its company resources to login attacks from remote non-employees			
1.	Consider case 1, as shown in table 1 , the suitable firewall type is _____			
	a) Screening routers	b) Stateful inspection	c) Application proxies	d) Guards
2.	Consider case 2, as shown in table 1 , the suitable firewall type is _____			
	a) Screening routers	b) Stateful inspection	c) Application proxies	d) Guards
3.	Consider case 3 , as shown in table 1, the suitable firewall type is _____			
	a) Personal firewall	b) Stateful inspection	c) Application proxies	d) Guards

Case 1	A school wants to allow its students to retrieve any information from www. resources on the internet. to help provide efficient service, the school wants to know what sites have been visited and what files from those sites have been fetched; particularly popular files will be cached locally.			
Case 2	University students and employees in offices use cable moderns or DSL connections with limited access.			
1.	Consider case 1 , as shown in table 1 , the suitable firewall type is _____			
	a) Screening routers	b) Stateful inspection	c) Physical separation	d) Application proxies
2.	Consider Case 2, as shown in table 1, the suitable firewall type is _____			
	a) Personal Firewall	b) Screening routers	c) Application Proxies	d) Guards

1.	A company wants to set up an <u>online price list</u> so that outsiders can see the products and prices offered, it wants to be sure that <u>no outsider can change the prices or product list</u> and <u>the outsider can access only the price list, not any of the more sensitive files stored inside</u> , the recommended firewall type will be _____			
	a) Proxy gateway	b) Application proxies	c) Guards	d) Personal firewalls
2.	A university wants to allow its students to use e-mail up to a limit of so many messages or so many characters of e-mail in the last so many days . Although this result could be achieved by modifying e-mail handlers, it is more easily done by monitoring the common point through which all e-mail flows, the mail transfer protocol. The recommended firewall type will be _____			
	a) Proxy gateway	b) Application proxies	c) Guards	d) Personal firewalls



MCQ PROBLEMS EXPLANATION

1. The result of encoding a plaintext "TECHNOLOGY" based on playfair cipher with key "SKY", consider Q is the less character

أي playfair له قواعد ثابتة:

هيديك اول حاجه ياما cipher تفكه ياما plaintext تشفره،

يمكن في السؤال ياما يديك الجدول جاهز ياما يديك متطلبات تعمل بيها الجدول اللي هي الKEY والLess character هنرسم جدول 5x5 هيبقى عندك 25 خلية والابجدية 26 حرف، هنكتب في اول كام خانة الkey اللي هو SKY هنكتبه في اول ثلاث خانات

بعد كدا هتكتب بقيت حروف الأبجدية من غير ما تكررهما وتكتب الless used character عشان الابجدية 26 حرف قولنا 😊

بعد ما نكون خلصنا خطوات تكوين الجدول هناخد الplaintext او الcipher نقسمه اتنين حروف زي كدا: **TE CH NO LO GY** وهنبداً ساعتها عملية من اتنين:

• عند تشفير الplaintext:

- بتتجه **يمين** لما الحرفين يبقوا في نفس الصف
- بتتجه **تحت** لما الحرفين يبقوا في نفس العمود
- لما الحرفين يبقوا مقابلين بعض في مستطيل اختار الزاويتين التانين زي TE تحت هتبقى PG

• عند فك التشفير بتعمل العكس:

- بتتجه **شمال** لما الحرفين يبقوا في نفس الصف
- بتتجه **فوق** لما الحرفين يبقوا في نفس العمود
- لما الحرفين يبقوا مقابلين بعض في مستطيل اختار الزاويتين التانين عادي زي اللي فوق

ملحوظات مهمة:

لو الplaintext فيه حرف مكرر بتحط بدال الحرف المتكرر X

لو عدد حروف الplaintext مش زوجي بتحط في الاخر ياما X ياما Q

لو لاحظت انك في خطوة الOR في التشفير انها مش OP فده عشان الP/Q دول less used character

TE CH NO LO GY
PG HN OR IR EB

S	K	Y	A	B
C	D	E	F	G
H	I	J	L	M
N	O	p/q	R	T
U	V	W	X	Z

- a) **PGHNORIREB** b) PGNUPRIREB c) QGNUGRIREB d) QGHUOQIREB

2. Consider the cipher text obtained from Q1, Encrypting it with row transposition algorithm, K=2, results

هنا عملنا صفين بما ان الk يساوي 2 وكتبنا اول حرف فوق ثاني حرف تحت ثالث حرف فوق رابع حرف تحت وهكذا...

P	H	O	I	E
G	N	R	R	B

- a) PNPIEGURRB b) **PHOIEGNRRB** c) QNGIEGURRB d) QGHUOQIREB

3. Based on the play fair matrix of Q1, the plaintext of the cipher text "RYTNEW" is _____

- a) Hello b) Sunset c) **Party** d) None



4.	Using the play fair matrix shown is fig 1 , the cipher of "FACULTY" will be				M	F	H	I/J	K
					U	N	Q	P	Q
					Z	V	W	X	Y
					E	L	A	R	G
					D	S	T	B	C
	a)	HLDQSAKX	b)	HDQASLXK	c)	LHQDSAKX	d)	HLDQASXK	

5.	Encrypting P = "stay home to be safe" by block transposition algorithm k = 32514, then C is												
لذقنا وقسمنا الplaintext بلوكات كل بلوك فيه 5 حروف بناء على عدد ارقام المفتاح k													
stayhometobesafe Stayh ometo besaf exxxx 12345													
ATHSY EMOOT SEFBA XXXEX 32514													
لو k = 32514 يبقى نستنتج ان: اول حرف هيبقى ثالث واحد في التشفير تاني حرف هيبقى ثاني حرف ثالث حرف هيبقى خامس حرف رابع حرف هيبقى اول حرف خامس حرف هيبقى رابع حرف													
P: STAYH 12345 --- C: ATHSY 32514 Ciphertext: athsy emot sefba xxxex													
a)	SOBETMEWAESXYTAJ HOFZ			b)	AESXTMEWHOFZSO BEYTAJ			c)	UOFZYTAAESXTME WSOBD		d)	None	

6.	In Caesar cipher, <u>shifting a letter by +3 positions</u> means that:			
	a)	A becomes D, B becomes E, C becomes F, and so on		
	b)	A becomes X, B becomes Y, C becomes Z, and so on		
	c)	A becomes C, B becomes F, C becomes I, and so on		
	d)	A becomes B, B becomes C, C becomes D, and so on		



7.	Based on the Playfair matrix of Fig. 1, the plaintext of the cipher text "NXOMTW" is _____ NX → PV OM → UH TW → AO Plaintext: PVUHAO					M	F	H	I/J	K
						U	N	O	P	Q
						Z	V	W	X	Y
						E	L	A	R	G
						D	S	T	B	C
						M	F	H	I/J	K
						U	N	O	P	Q
						Z	V	W	X	Y
						E	L	A	R	G
						D	S	T	B	C
a)	Hellox	b)	Sunset	c)	Party	d)	None			

8.	Encrypting P= "COMPUTER NETWORK" by Block transposition algorithm k=32514, then C is: COMPU TERNE TWORK COMPU 12345 MOUCP 32514 MOUCP RETN OWKTR								
a)	PNROEWCTTUEKMR O	b)	TWORKTERNECOMP U	c)	CTTOEWMROPNRU EK	d)	None		

9.

Encrypting p = "computers security" using **row transposition** algorithm K = 425163 , then C is _____

بناء على عدد ارقام المفتاح هتحدد عدد العواميد اللي هما 6 هنا

In row transposition write the plaintext in 6 columns

رتب العواميد تصاعدي

Then rearrange the columns ascendingly

بعدها اكتب عمود بعمود

C4	C2	C5	C1	C6	C3
C	O	M	P	U	T
E	R	S	S	E	C
U	R	I	T	Y	X

C1	C2	C3	C4	C5	C6
P	O	T	C	M	U
S	R	C	E	S	E
T	R	X	U	I	Y

Ciphertext: **PSTORRTCXCEUMSIUEY**

a)	PEYORIUCJERTUQ MST	b)	CERORIMSTPEYUCJT UQ	c)	PEYUCJTUQMSTORI CER	d)	None
----	-----------------------	----	------------------------	----	------------------------	----	------

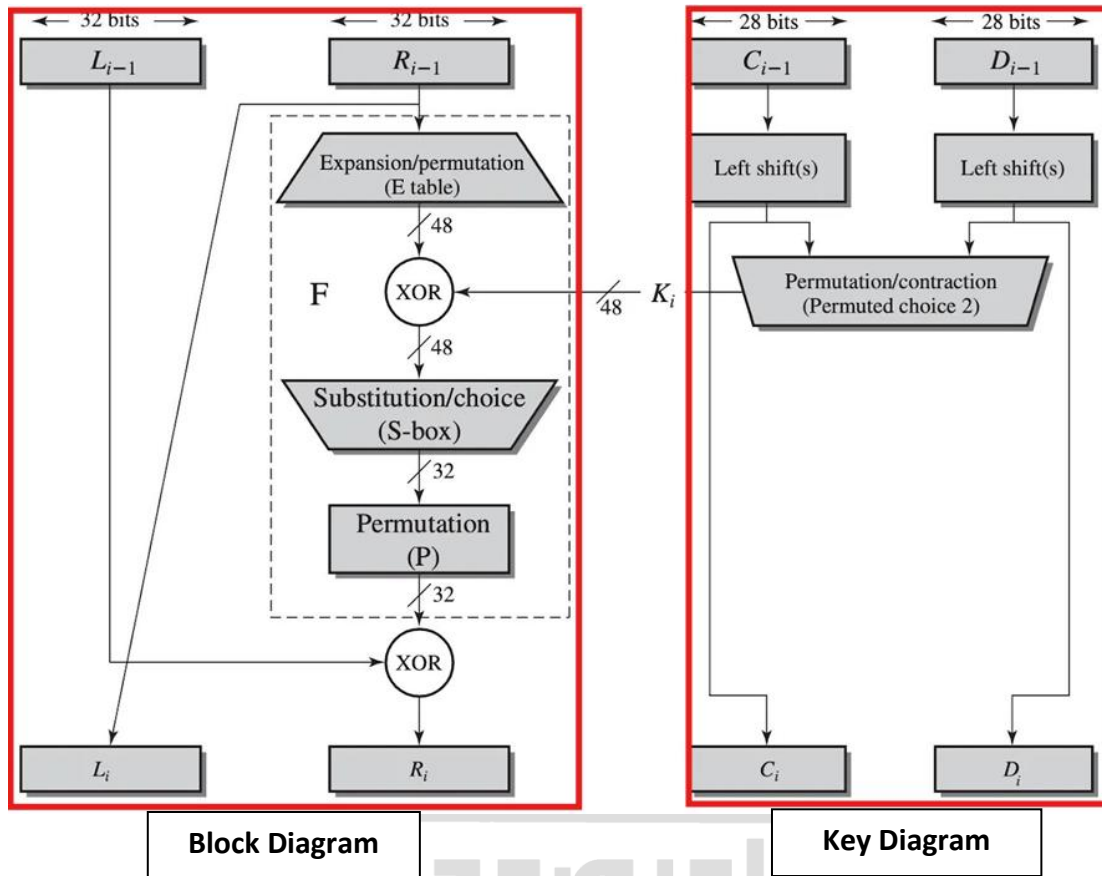


WRITTEN EXPLANATION

DES Single Round

حجم الblock: 64-bit
حجم الkey: 56-bit (Single DES)
عدد الدورات (rounds): 16 دورة

احنا عندنا في ال single round جزئين جزء key diagram وجزء block diagram



بالنسبة للkey:

احنا عندنا ناحيتين c و d اتعملوا عشان نقسم الkey ونعمل left shift لكل نص وبعدين permutation وفي الخطوة دي بنحذف الأصفار الزيادة من الkey فهيطلع 48-bit key (K_i)

بالنسبة للblock:

احنا عندنا في ال block الداتا بتدخل في ال block تنقسم لجزئين left و right كل جزء فيهم بيبقى 32-bit اول خطوة بتحصل هي ان اصل ال right يروح لل left وبعدين ال right هيدخل على حاجة اسمها ال Feistel function (F-function)



هنشرح كل خطوة في ال-f-function:

1- Expansion (E table):

بتحصل في الخطوة دي امتداد للblock اليمين بحيث يبقى من 32-bit الى 48-bit

2- XOR with round key:

في المرحلة دي بيكون عملنا shifting للkey وبتبقى القيمة بتاعته 48-bit

3- Substitution (S-box):

بيرجع القيمه من 48-bit إلى 32-bit

4- Permutation (P-box):

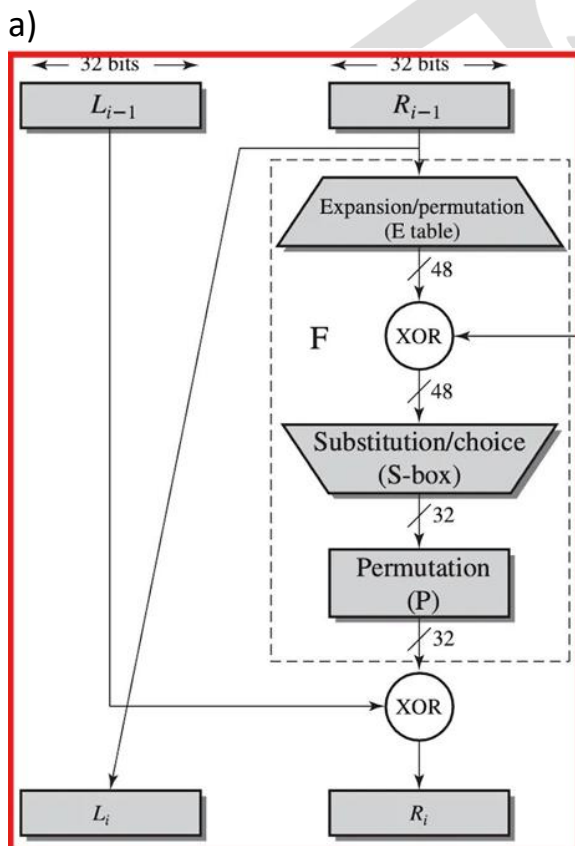
ببديل مخارج الbits

بعد ما يخرج من ال-f-function هيعمل XOR للleft مع اللي خارج من ال-F-function

وهنا بييجي سؤال محتمل انك ترسم block diagram

a) Draw block diagram of a single round in DES

b) Mention the steps of F-function



b) Expansion, XOR with round key, Substitution, Permutation



الاحتياط واجب

Choose the correct answer:

1.	Ensuring that computer assets are accessed only by authorized parties and authorized parties actually access the required assets is a goal of _____.			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
2.	_____ is Ensuring that the data is accurate, consistent, and not tampered with .			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
3.	Using systems like Git to track and prevent unapproved modifications in codebases achieves _____ of the data			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
4.	_____ Is Ensuring that the systems and the data are accessible when needed .			
	a) Confidentiality	b) Integrity	c) Availability	d) All of the above
5.	Service is <u>denied to unauthorized users</u> is an example of data availability (confidentiality)			
	a) TRUE	b) FALSE	c)	d)
6.	When an asset of the system is destroyed or becomes unusable , this is called _____ and this is an attack on the _____			
	a) Interruption, availability	b) Interception, availability	c) modification, availability	d) modification, integrity
7.	An unauthorized party only gains access to an asset this is called _____ and this is an attack on the _____.			
	a) Interruption, availability	b) Interception, confidentiality	c) modification, availability	d) modification, integrity
8.	An unauthorized party not only gains access to an asset but tampers with an asset this is called _____ and this is an attack on the _____.			
	a) Interruption, availability	b) Interception, confidentiality	c) modification, availability	d) modification, integrity
9.	“ Who are you ”? is an _____ question while “ What are you allowed to do ?” is an _____ question			
	a) Authentication, Authorization	b) Authorization, Authentication	c) Authentication, Authentication	d) Authorization, Authorization

